

Welcome!





Who Am I?



Python Introduction!



 *“Let’s unlock the door to programming with Python!”*

What is Python?

Python is a **high-level, interpreted, general-purpose programming language** known for its **simplicity** and **readability**.

History:

- Developed by **Guido van Rossum**
- Released in **1991**
- Named after the TV show “*Monty Python’s Flying Circus*”, not the snake 

Visual: Timeline with a Python snake, computer icon, TV symbol

 *Think of Python as a language that talks to your computer – simply and clearly.*



➤ Why Learn Python?

Reasons Why Python is Popular:

- Easy to Learn – Simple syntax like English
- Versatile – Web, Data, ML, AI, Automation
- High Demand – Top skill in tech jobs
- Large Community – Help is always available
- Huge Libraries – Ready-made tools for every task

🎯 Visual Idea:

Pie chart with areas of usage → Web 🕸, Data 📊, AI 🤖, Automation ⚙

🧠 *Python lets you focus on problem-solving, not syntax memorization.*



➤ Where is Python Used?

Real-world Applications:

-  **Data Science** – NumPy, Pandas, Matplotlib
-  **Machine Learning / AI** – TensorFlow, Scikit-learn
-  **Web Development** – Django, Flask
-  **Automation & Scripting** – Selenium, OS scripts
-  **Game Development** – Pygame
-  **Cybersecurity, IoT, App Development, etc.**
-  **Example Companies Using Python:**

Netflix, Google, NASA, Facebook, YouTube

 *Python is everywhere – from satellites to search engines.*



➤ Python is Beginner Friendly

🧠 Python syntax is clean and simple. Example:

Python

```
print("Hello, world!")
```

Compare this to C:

C

```
#include <stdio.h>
int main() {
    printf("Hello, world!");
    return 0;
}
```

🚀 Python makes coding faster and easier for beginners.

🧠 **Tip:** Start small, build big – Python scales with you.



➤ Python Features at a Glance

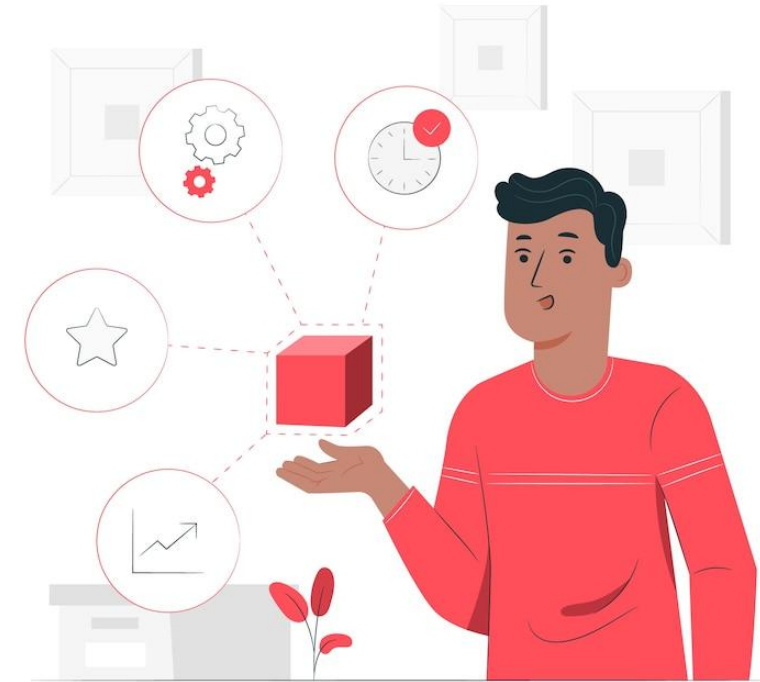
Key Features:

-  Interpreted Language
-  Easy to Learn & Use
-  Portable & Cross-Platform
-  Large Standard Library
-  Supports OOP and Functional Programming
-  Dynamically Typed (no need to declare variable types)

Example:

python

```
x = 5           # Integer  
x = "Hello"     # Now it's a string
```



 Python grows with your ideas – one line at a time.



➤ Installing Python

Steps:

1. Visit  <https://www.python.org>

2. Go to **Downloads**

3. Choose Python 3.x (Latest Stable Version)

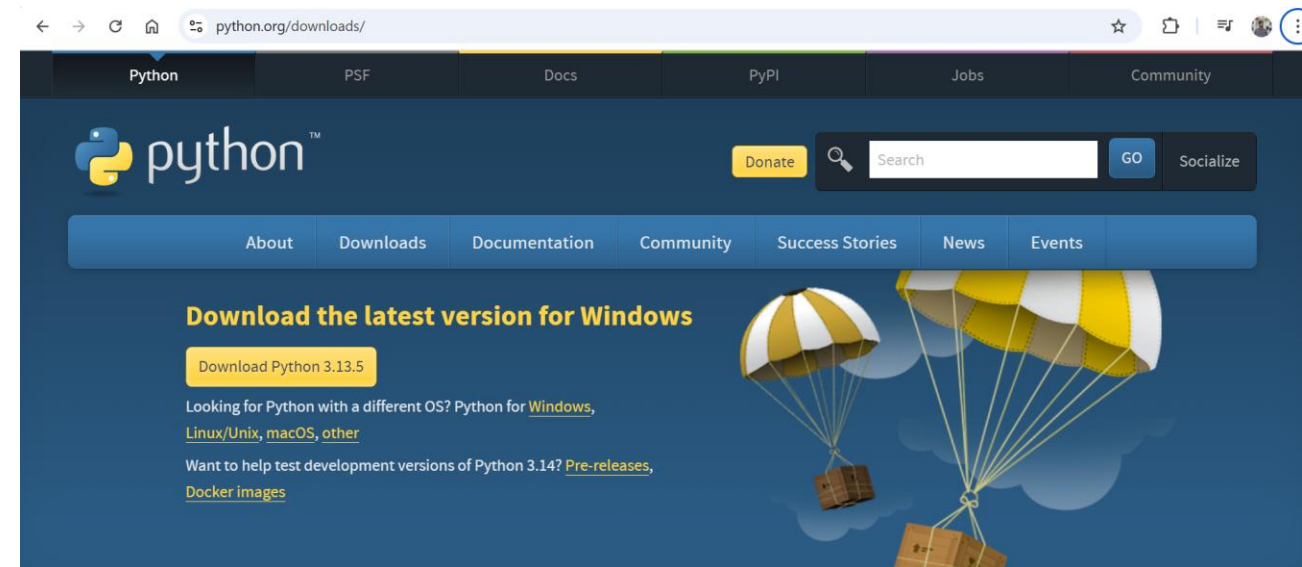
4. Install using default settings

5. Check version:

bash

```
python --version
```

 *Always use the latest version unless your project needs older one.*



➤ Ways to Run Python Code

Options:

1. **IDLE (Default):** Comes with Python
2. **VS Code:** Modern editor with autocomplete and extensions
3. **PyCharm:** Best for large Python projects
4. **Online Editors:**
 1. Replit
 2. Google Colab
 3. Jupyter Notebook
 4. Programiz

 *Choose what fits your comfort level – all roads lead to Python!*



➤ Python File Extension

📁 Python files use .py extension

✅ Examples:

- main.py
- calculator.py
- hello_world.py

📦 To run from command line:

Bash: `python hello_world.py`

💡 *.py = Python program. Save all your scripts this way.*



